

AI-Powered Automatic Data Analysis

(Smart analytics for smarter decisions)

National Institute of Electronics and Information Technology



SRINAGAR CAMPUS

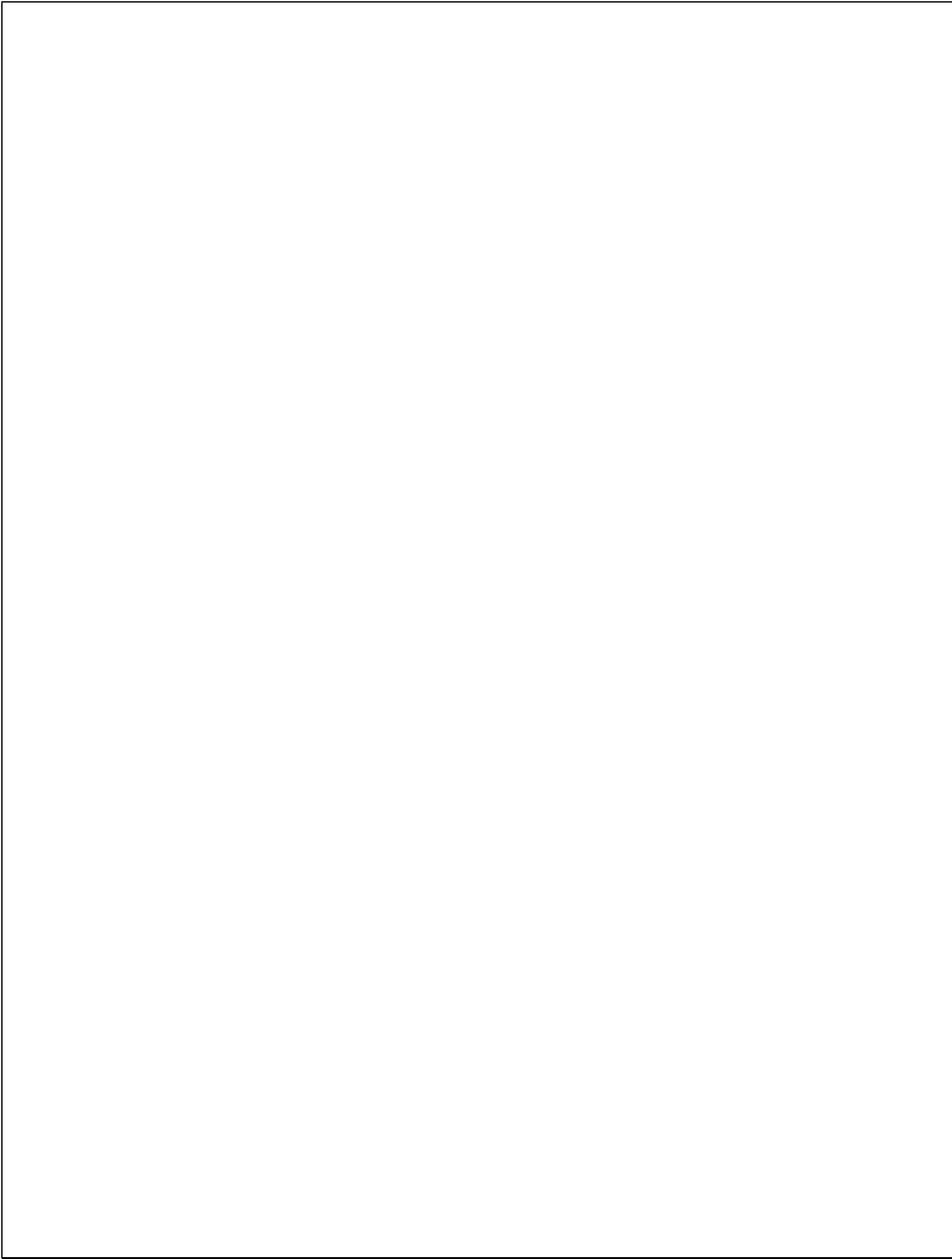
**SIDCO Electronics Complex Rangreth Old Airport,
Srinagar, Kashmir**

Affiliated to

University of Kashmir



Department of Computer Science



AI-Powered Automatic Data Analysis

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PROJECT REPORT

Submitted by

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in partial fulfillment for the award of the degree

of

MCA (Masters of Computer Applications)

in

Department of Computer Science

University of Kashmir

National Institute of Electronics and Information Technology

BONAFIDE CERTIFICATE

Certified that this project report titled

“AI-Powered Automatic Data Analysis”

is the bonafide work of Mr. Waqar Maqbool and Mrs. Inshpreet Kaur Mehta, who
carried out the project work under my supervision and guidance in the **Department of**
Computer Science during the academic year **2025**.

This work has not been submitted previously for the award of any degree, diploma, or
certificate at any other university or institute.

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DECLARATION

We hereby declare that the project report titled “**AI-Powered Automatic Data Analysis**” is an original work carried out by us as a part of our academic curriculum in the **Department of Computer Science**.

This work has been conducted under the guidance of **Dr. Syed Nisar Bukhari (Scientist-D)**, and we affirm that it has not been submitted previously to any other university or institution for the award of any degree, diploma, or certificate.

We also confirm that the content presented in this report is the result of our own efforts and research. Proper acknowledgments have been given wherever the work of others has been cited.

We understand that any form of plagiarism or misrepresentation may lead to disciplinary action as per institutional policies.

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ACKNOWLEDGEMENTS

We are sincerely thankful to Almighty God for blessing us with the strength, patience, and wisdom to successfully complete this project. Without His guidance and grace, this work would not have been possible.

We also extend our deepest gratitude to our parents, whose constant love, encouragement, and unwavering support have been our greatest source of motivation throughout this journey.

We would like to express our heartfelt appreciation to Dr. Syed Nisar Bukhari, Project Supervisor, NIELIT Srinagar, for his exceptional mentorship and invaluable guidance. His expertise in Machine Learning and Artificial Intelligence played a pivotal role in shaping the direction of this project.

Our sincere thanks to Er. Sarfaraz Sir, Software Development Incharge, NIELIT Srinagar, for his technical advice, problem-solving insights, and continued encouragement that greatly enhanced our project's execution.

We are also grateful to Dr. Basab Nath, Assistant Professor, IILM University, for his inspiring academic insights into Natural Language Processing, which helped broaden our understanding and contributed to the AI aspect of our system.

We would like to thank the Department of Computer Science NIELIT, for providing a collaborative and resourceful environment that made this work possible.

A heartfelt thanks to our classmates and friends for their moral support, discussions, and constructive feedback during the development phase.

This project has been a valuable learning experience, and we are deeply thankful to everyone who contributed to its successful completion.

Waqar Maqbool Wani

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ABSTRACT

In the big data age, it is still very much a lengthy and laborious process of transforming raw datasets into meaningful insights, particularly for non-technical users. This project, AI-Powered Automatic Data Analysis undertakes to bridge that gap with an easy, web-based system whereby users can upload their datasets and then conduct full data analysis through Natural Language Processing (NLP).

The system is built using Python and Flask, with Pandas to do the main data processing work. It allows data upload in several formats- CSV, Excel, JSON, and it will immediately do cleaning, summarization, anomaly detection, and visualization. A big part of the platform is an AI-powered chatbot included through OpenRouter API which helps users talk to their data using simple English questions. This makes big analysis work like filtering, adding up and finding patterns easier without having to write code.

Users can explore the data through an interactive dashboard, generate charts, and apply filters in real-time. There is also a feedback and review module to collect user inputs for helping to improve the tool iteratively.

The architecture has been designed modular and extensible to enable the easy integration of future Machine Learning tools and services. It aims at democratizing data analytics, making it accessible to a wider audience, and more importantly, non-technical backgrounds.

This report documents the system's design, implementation, key features, and potential future enhancements. It presents an in-depth view of how Artificial Intelligence can simplify as well as improve the analysis of data.

Table of Contents

Chapter 1 Introduction

- 1.1 Project Overview
- 1.2 Problem Statement
- 1.3 Objectives
- 1.4 Scope of the Project
- 1.5 Motivation and Significance
- 1.6 Organization of the Report

Chapter 2 Literature Review

- 2.1 Data Analytics and Automation
- 2.2 Natural Language Processing in Data Analysis
- 2.3 Review of Existing Platforms
- 2.4 Identified Gaps and Project Uniqueness
- 2.5 Overview of Technologies Used

Chapter 3 System Analysis and Design

- 3.1 System Requirements (Functional & Non-Functional)
- 3.2 High-Level Architecture
- 3.3 Component-Level Breakdown
 - a. Data Upload & Preprocessing
 - b. NLP Chatbot Integration
 - c. Visualization Engine
 - d. Dashboard Filtering/Sorting
 - e. User Reviews

- 3.4 Database Design (SQLAlchemy schema for feedback)
- 3.5 UI Design and User Flow

Chapter 4 Implementation

- 4.1 Project Setup and Environment
- 4.2 Backend Logic (main.py and Flask Routing)
- 4.3 Data Processing (Pandas logic)
- 4.4 Chatbot & NLP Layer (OpenRouter API)
- 4.5 Frontend Templates and Dynamic Rendering
- 4.6 Directory & Codebase Structure
- 4.7 Error Handling & Edge Case Management

Chapter 5 Testing and Evaluation

- 5.1 Testing Strategy
- 5.2 Unit and Integration Test Cases
- 5.3 Chatbot Input Scenarios
- 5.4 UI/UX Feedback
- 5.5 Performance Metrics

Chapter 6 Results and Discussion

- 6.1 Upload and Preview Demo
- 6.2 Data Summary Generation
- 6.3 Interactive Filtering and Visualization
- 6.4 AI Chatbot Query Examples and Accuracy
- 6.5 Sample User Feedback

6.6 Effectiveness and Limitations

Chapter 7 Future Enhancements

7.1 Additional ML Tools and Modules

7.2 User Authentication and Session Storage

7.3 Dataset Annotation and Export

7.4 Deployment Optimization

7.5 Mobile Responsive Version

Chapter 8 Conclusion

8.1 Summary of Achievements

8.2 Key Takeaways

8.3 Project Impact

Chapter 9 References

Chapter 10 Appendices

A. Sample Dataset Format

B. Screenshot Gallery

C. NLP Prompt Engineering Samples

D. Setup and Deployment Guide

E. Important Code Snippets